

Nguyen Dao Nam Hai

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CAREER OBJECTIVE

Orientation: Edge AI & Computer Vision Engineer

Eager to leverage a strong software engineering foundation and expertise in AI, Embedded Systems, and IoT to develop intelligent Edge AI solutions. My goal is to build autonomous vision systems, optimize AI Agents and computer vision models for resource-constrained devices, and design scalable, low-latency architectures for real-time multimodal intelligence.

EDUCATION

Hanoi University of Science and Technology (HUST)

Major: Information Technology

Program: Computer Engineering (IT2)

Hanoi, Vietnam

Sep 2023 – Present

CPA: 3.31/4.0

Bien Hoa High School for the Gifted

Chemistry Major (K62)

Ha Nam, Vietnam

Sep 2020 – Jun 2023

TECHNICAL SKILLS

- **Programming:** Python, Go, C/C++ , Java, Kotlin, SQL, JavaScript, HTML/CSS.
- **AI & Computer Vision:** TensorFlow/Keras, OpenCV, Google Gemini API, TinyML.
- **Backend:** Flask, SQLAlchemy, GORM, RESTful APIs, JWT Authentication, Microservices.
- **Frontend:** React, JavaScript, HTML5/CSS3.
- **Databases:** PostgreSQL, InfluxDB, SQLite.
- **Embedded & Tools:** ESP32, STM32, MQTT, TouchGFX, Git, Linux (Ubuntu), Agile.

WORK EXPERIENCE

Piscada Software Vietnam | Software Developer Intern

Feb 2026 – Present

- Developed backend services in Python for a real-time industrial platform based on Microservices architecture.
- Designed and optimized hybrid data storage using PostgreSQL and InfluxDB for structured and time-series data.
- Built interactive React dashboards and integrated MQTT with RESTful APIs for real-time monitoring and device communication.

VinUniversity | AI Practical Talent Program – Cohort 3

Jul 2026 – Present

- Selected as a trainee in Cohort 3 of VinUniversity's AI Practical Talent Program.
- Receiving intensive training in Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), AI system design, and production-ready AI application development.

Edge AI Laboratory, SOICT | Research Assistant

Feb 2026 – Aug 2026

- Developed Edge AI inference pipelines and implemented bidirectional communication between embedded sensor nodes and monitoring dashboards.
- Optimized computer vision and AI inference algorithms for resource-constrained embedded platforms, improving real-time execution efficiency.

PERSONAL PROJECTS

Finance-Agent (AI SaaS Personal Finance Platform) | Go, Gemini API, SQLite, GORM, JWT, JavaScript

- **High-Concurrency Backend:** Architected a scalable SaaS platform using Goroutines, Channels, asynchronous Job Queues, API Key Rotation, and Token Bucket Rate Limiting to support high-concurrency AI workloads while preventing API quota exhaustion.
- **LLM Integration & System Design:** Integrated Google Gemini Multimodal API to extract structured financial transactions from natural-language conversations and receipt images. Developed secure JWT authentication, real-time financial dashboards, and responsive client-side image compression for efficient AI inference.

AI-Driven Numerical IV Drip Hub | Python, ESP32, TinyML, MQTT

- **Edge AI Healthcare System:** Designed an intelligent IV monitoring system integrating numerical methods (Bisection, Newton-Raphson) with continuous sensor calibration for accurate flow-rate estimation.
- **TinyML Deployment:** Deployed TinyML models on ESP32 for real-time anomaly detection and established MQTT-based communication between embedded devices and monitoring dashboards with low-latency data transmission.

GROUP PROJECTS

BlueMoon Apartment Management System | Frontend Lead | React, TailwindCSS, Axios

- **Frontend Architecture:** Led the development of responsive administrative dashboards for apartment management, optimizing complex data visualization, billing, and maintenance workflows.
- **Team Leadership & Integration:** Coordinated frontend development across 20+ functional modules, conducted code reviews, and integrated secure authentication and RESTful APIs using Axios.

Dynomite STM32F429I-DISCO | Embedded Software Developer | C/C++, STM32, TouchGFX, DMA, I2S

- **Embedded Graphics Engine:** Developed a 60 FPS embedded game on STM32F429I-DISCO using TouchGFX, MVP architecture, DMA2D double buffering, and optimized SDRAM memory management for real-time rendering.
- **Real-Time Audio & Algorithms:** Built a DMA-driven I2S audio engine with ADPCM decoding and software audio mixing. Optimized gameplay using BFS-based Match-3 detection, orphan-drop algorithms, and efficient collision computation for resource-constrained embedded hardware.

CERTIFICATIONS & ACHIEVEMENTS

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| • Finalist – USAII Global AI Hackathon | Jun 2026 |
| • Top 14 Finalist – SEAQuantathon International Quantum Computing Hackathon | Jul 2026 |
| • Samsung Innovation Campus Corporate Scholarship | 2025, 2026 |
| • Introduction to Internet of Things – Cisco Networking Academy | Mar 2026 |
| • Google Career Certificates: AI Essentials & Prompting | Dec 2025 |
| • "Student of 5 Merits" & Youth Union Merit (University Level) | 2023–2024, 2024–2025 |

EXTRACURRICULAR ACTIVITIES

- **Vice Captain of Volunteer Squad, HUST Student Association** (Oct 2023 – Dec 2025): Managed personnel for large-scale campaigns and developed teamwork skills in diverse environments.